CWI Centrum Wiskunde & Informatica

On the Composition of 2-Prover Commitments and Multi-Round Relativistic Commitments

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Warning

- My talk:
  mainly classical



- Still interesting for you:
  - topic (relativistic commitments)
  - some of our results hold in quantum setting
  - interesting open questions in quantum setting

- Will use:

 and 

to distinguish between classical and quantum

Commitment Schemes

- Commitment scheme = digital analogue of:
  - putting something in a vault
  - and revealing it later





- Security properties:
  - hiding: verifier cannot see value “inside” commitment
  - binding: prover cannot change his mind

- Important cryptographic primitive, used for
  - coin tossing
  - zero-knowledge proofs
  - multiparty computation
  - etc.

(Im)possibility

- Well known / easy to see:

  scheme **cannot** be **unconditionally** hiding and binding

- Even quantum communication/computation does not help

- Need to restrict one of the parties:
  - computationally-bounded prover (or verifier)
  - prover (or verifier) with bounded (quantum) memory
  - two-prover setting with non-communicating provers

[BenOr,Goldreich,Kilian,Wigderson 1988]

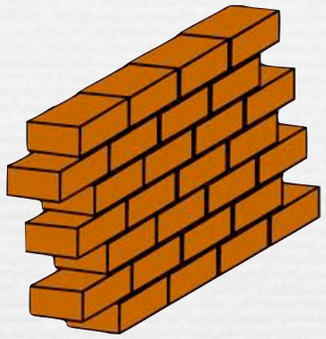
Example Two-Prover Bit Commitment Scheme

[Crépeau, Salvail, Simard, Tapp 2011]

$b \in \mathbb{F}_2 \ , \ \ y \in \mathbb{F}_2^n$

PROVER P





PROVER Q



commit

$a \in \mathbb{F}_2^n$

$x := y + ab$

VERIFIER V



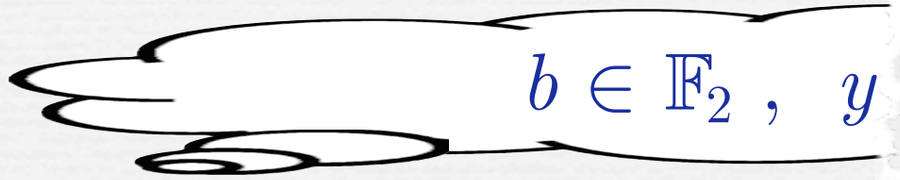
y

open

$x \oplus y \stackrel{?}{=} ab$

Example Two-Prover Bit Commitment Scheme

[Crépeau, Salvail, Simard, Tapp 2011]



PROVER P



$a \in \mathbb{F}_2^n$

$x := y + ab$

VERIFIER Q



$x \oplus y \overset{?}{=} ab$

Security:

- Hiding: obvious
- Binding:
  - opening $b=0 \Rightarrow y=x$
  - opening $b=1 \Rightarrow y=x \oplus a$
  - both $\Rightarrow$ Q knows a



- Binding:
  - does still hold
  - proof is more involved
  - square-root loss



Relativistic Commitment Schemes

[Kent 1999, 2005 etc.]

In essence:

relativistic commitment scheme

=

2-prover commitment scheme

+

assumption enforced by special relativity

Example Relativistic Bit Commitment Scheme

$b \in \mathbb{F}_2 \ , \ \ y \in \mathbb{F}_2^n$

PROVER P



PROVER Q



$a \in \mathbb{F}_2^n$

$x := y \oplus ab$

VERIFIER V



$x \oplus y \overset{?}{=} ab$

commit

y

open

Outside of the light cone!

Example Relativistic Bit Commitment Scheme

Disadvantage:

open must take place veeeeeery shortly after *commit*





$a \in \mathbb{F}_2^n$

$x := y \oplus ab$

VERIFIER V



$x \oplus y \stackrel{?}{=} ab$

commit

open

y

Outside of the light cone!

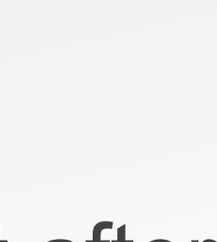
Multi-Round Schemes





Security requirements:

- Hiding: until *open*
- Binding: from right after *commit*







A Multi-Round Scheme

[Lunghi, Kaniewski, Bussières, Houlman, Tomamichel, Wehner, Zbinden 2014]



$b \in \mathbb{F}_2 \,, \quad y_0, y_1, \ldots, y_m \in \mathbb{F}_2^n$



commit

$a_0 \in \mathbb{F}_2^n$

$x_0 := y_0 \oplus a_0 b$

$a_1 \in \mathbb{F}_2^n$

$x_1 := y_1 \oplus a_1 y_0$

$\mathbb{F}_2^n \approx \mathbb{F}_{2^n}$

sustain

$a_2 \in \mathbb{F}_2^n$

$x_2 := y_2 \oplus a_2 y_1$

$\vdots$

compute $y_{m\text{-}1}, \ldots, y_0$
check $x_0 \oplus y_0 = ab$



y_m

open

Previous Analysis, and Our Results

Lunghi et al. show:

- Theoretical analysis: m-round scheme is ε-binding with

$$\varepsilon \lesssim 2^{-\frac{n}{2^m}}$$

doubly exponential

- Practical implementation:
  can keep commitment alive for 2 ms.

We show:

- New analysis: m-round scheme is ε-binding with

$$\varepsilon \lesssim m \cdot 2^{-\frac{n}{2}}$$

$\Rightarrow$ Using data from their implementation:
could keep commitment alive for **10^{50} years!!!!**

Previous Analysis, and Our Results

Actually,

- main result: general "**composition theorem**" ()
- based on: **better understanding** of binding property ()
- improved security analysis **follows as corollary** ()

Independent work (Chailloux, Chakraborty & Leverrier):

- Same improved bound on Lunghi et al. scheme
- Tailored to the specific scheme
- Much less general / less insightful

Road Map

1. Intro

2. The Intuition

3. Defining the Binding Property

4. The Composition Theorem

5. Putting Things together

6. Wrapping Up

The Lunghi et al 2-Round Scheme

commitment to b

$a_0 \in \mathbb{F}_2^n$

$x_0 := y_0 \oplus a_0 b$

syntactically: commitment to y_1

$\mathbb{F}_2^n$

$x_1 := y_1 \oplus a_1 y_0$

syntactically: commitment to y_0

Sustain

$a_2 \in \mathbb{F}_2^n$

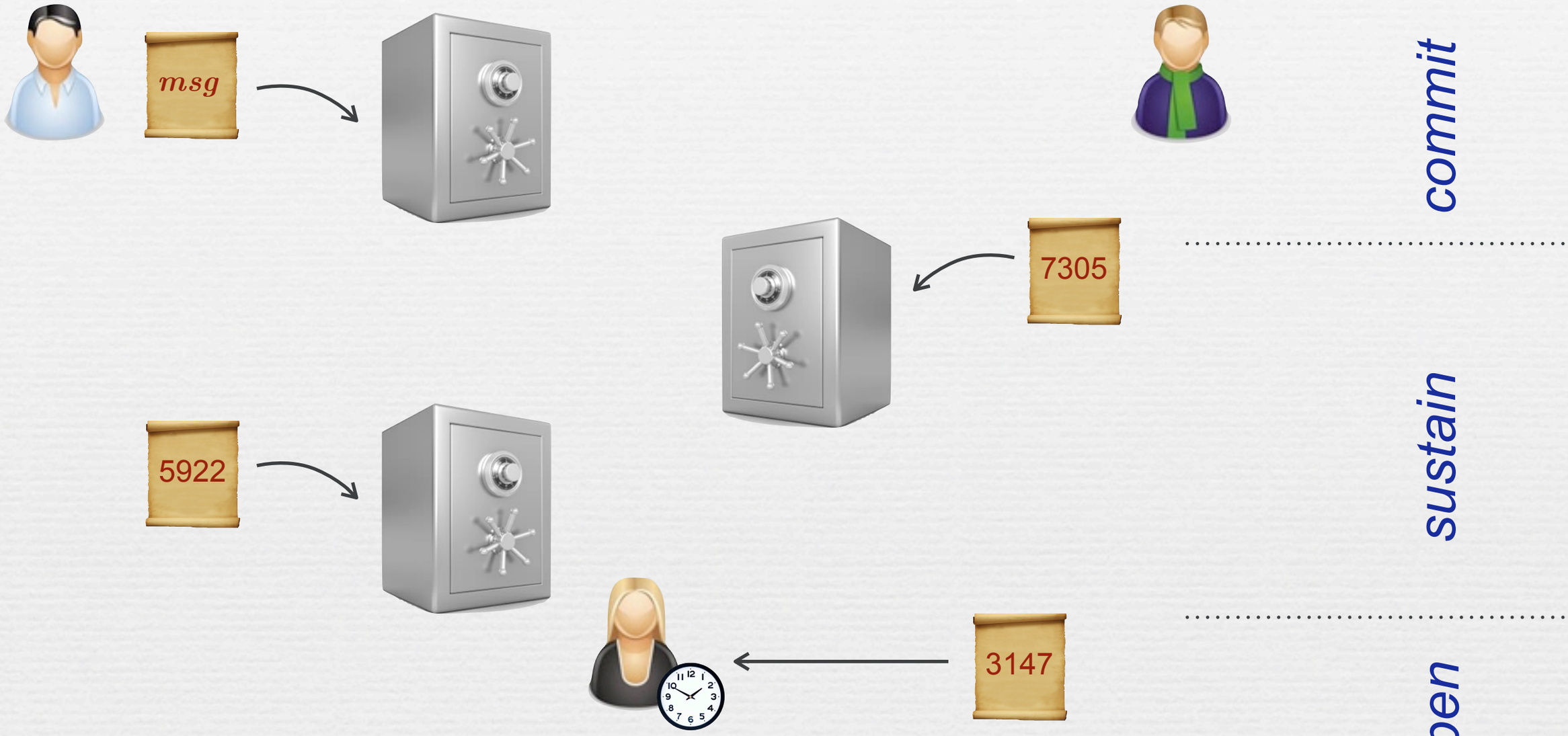
$x_2 := y_2 \oplus a_2 y_1$

y_2

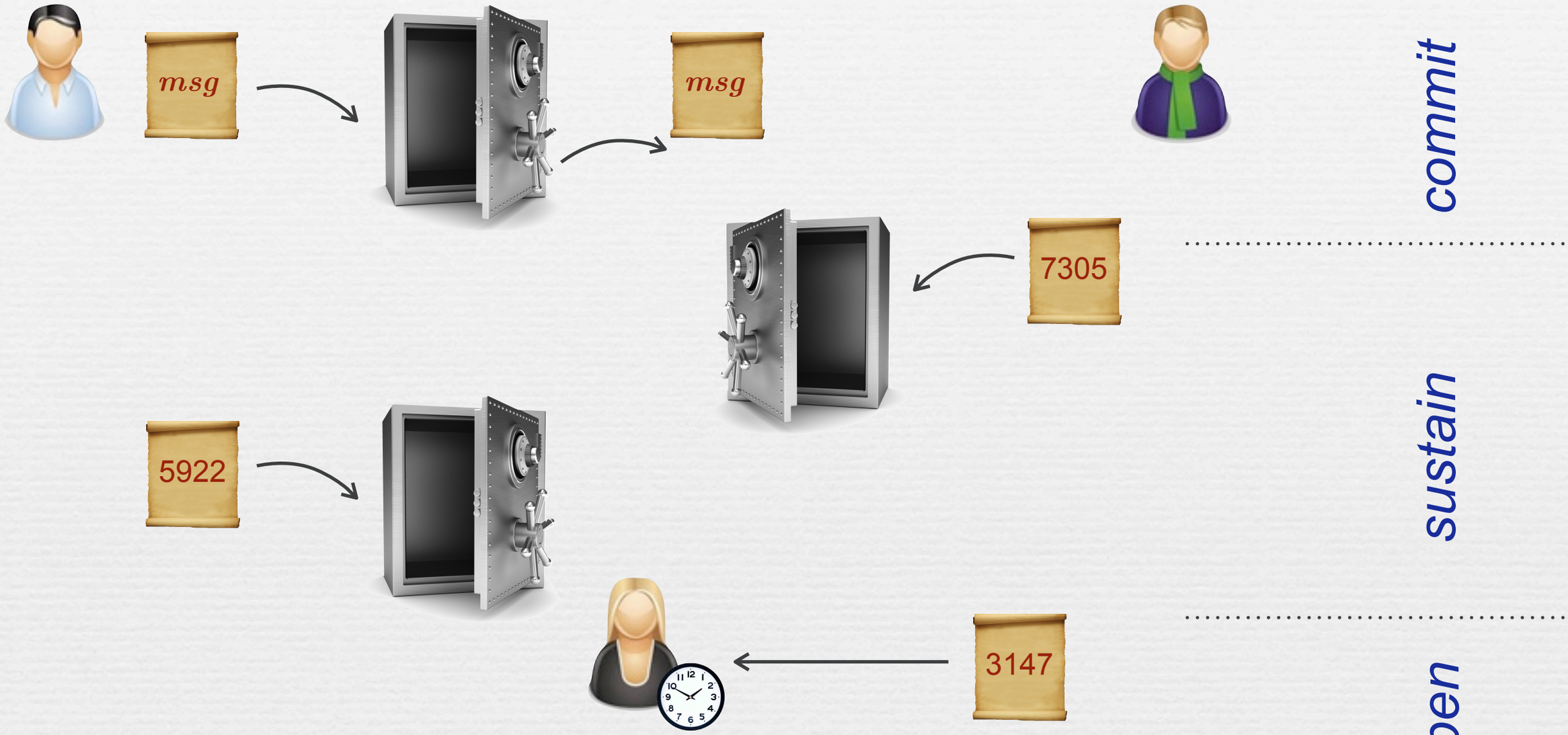
opens commitment to y_1

Open

The Lunghi et al Scheme - Schematically



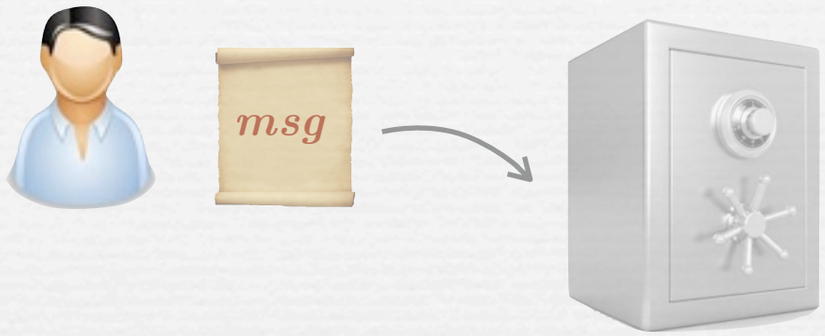
The Lunghi et al Scheme - Schematically

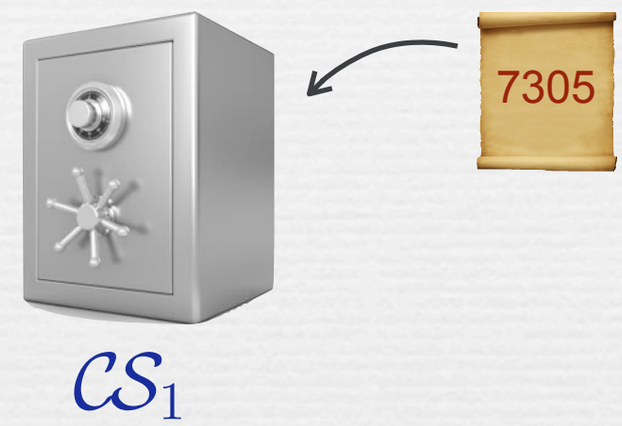


Q: Is this **composition of commitments** still **binding**?

Subtle issue: in *open*, provers could cheat original commitment.
Intuition: too late now, they're now committed to the opening

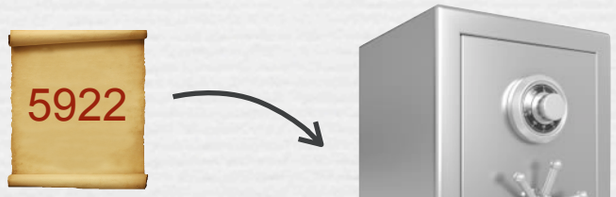
Composition of Commitments





$\mathcal{CS}_1$

$=: \mathcal{CS}_1 * \mathcal{CS}_2.com$



$\mathcal{CS}_2$



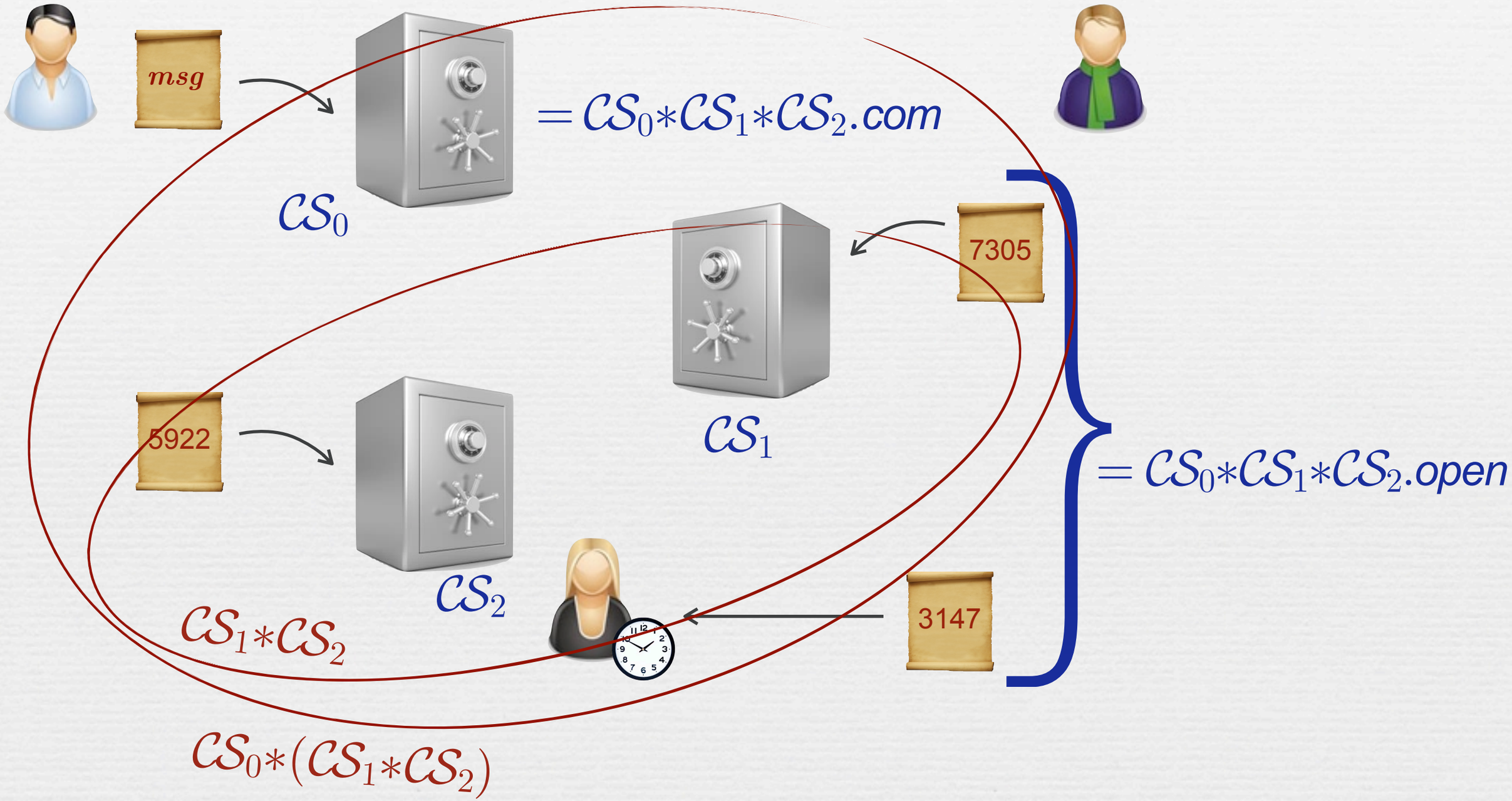


$=: \mathcal{CS}_1 * \mathcal{CS}_2.open$

Composition Theorem:

If $\mathcal{CS}_1$ & $\mathcal{CS}_2$ are ε- & δ-binding, then $\mathcal{CS}_1 * \mathcal{CS}_2$ is $(\varepsilon+\delta)$-binding.

Security of the Lunghi et al Scheme



By repeated application of the composition theorem:
security of Lunghi et al scheme **with linear blow-up**

Hurdles

1. Standard definition: “unfriendly” for proving composition

- Our solution:
  - introduce **new yet equivalent** definition
  - show composability by means of this new definition
  - (may be of independent interest)

2. *TBA*

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Defining the (Info-Theoretic) Binding Property

Notation:

- $s \in \{0,1\}^n \cup \{\perp\}$ = string accepted by verifier
- $p(s)$ = its distribution
- in case of a **bit** commitment: $b \in \{0,1,\perp\}$ and $p(b)$

Intuition: After committing, $\exists\, \hat{s} \in \{0,1\}^n$ so that for any *open*

$$s = \hat{s} \;\vee\; s = \perp$$

except with small probability.

Definition. A (2-prover) string commitment is ε-*binding* iff

$$\forall \textit{commit}\; \exists\, p(\hat{s})\; \forall \textit{open}\; \exists\, p(\hat{s}, s):\; p(\hat{s} \neq s \neq \perp) \leq \varepsilon$$

Remark: Is **not** the strongest possible definition.

Equivalence to Standard Definition

Theorem. A (2-prover) **bit** commitment is ε-binding iff

$$\forall \mathit{commit}\ \forall \mathit{open}_0, \mathit{open}_1 \colon\ p(b_0 = 0) + p(b_1 = 1) \leq 1 + 2\varepsilon$$

= common definition, as used by [Crépeau et al.], [Kent], [Lunghi et al.], etc.

Cor. The Crépeau et al. **bit** commitment is 2^{-n}-binding.

Cor. The Crépeau et al. **bit** commitment is $2^{-n/2}$-binding.

Recall: also need security as **string** commitment...

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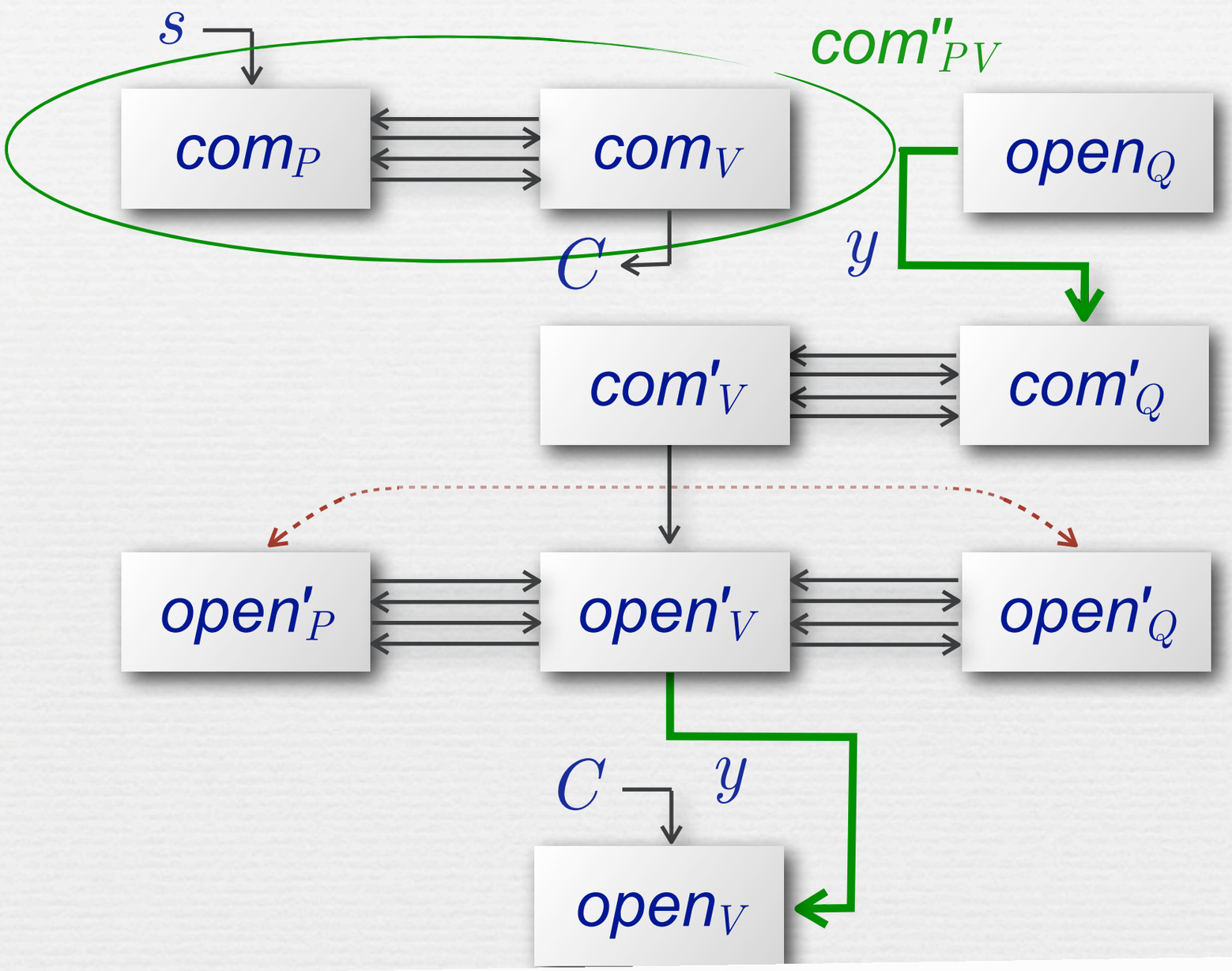
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Composing 2-Prover Commitments





$\mathcal{CS}$

$\mathcal{CS}'$

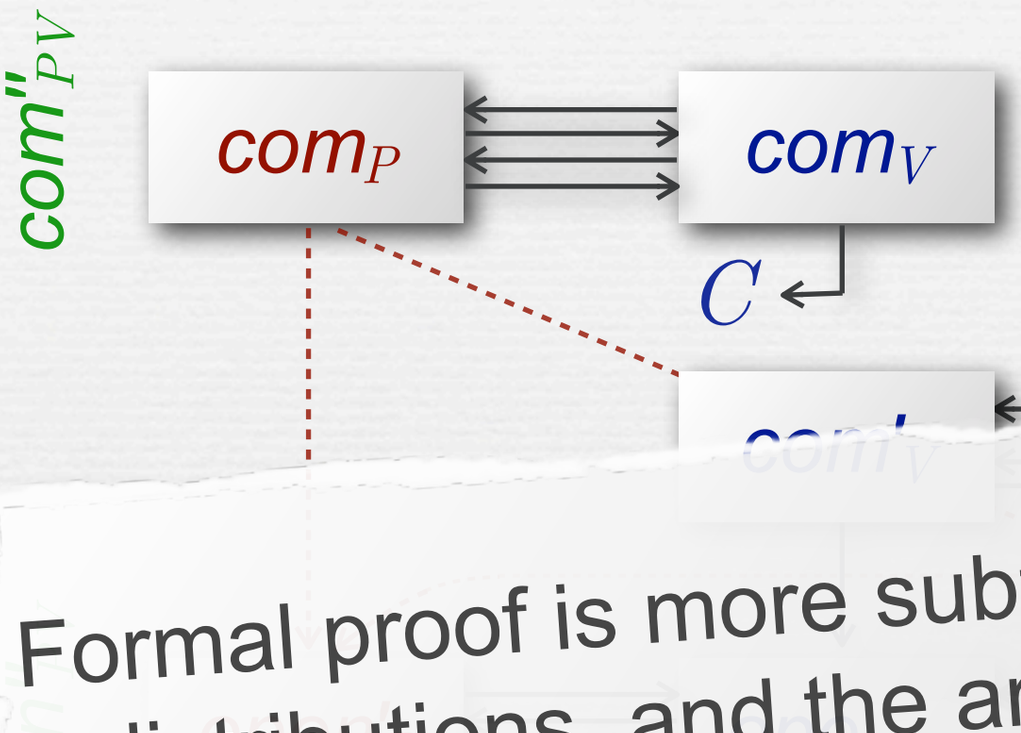
$open''_{PQV}$

$\mathcal{CS}'' = \mathcal{CS} * \mathcal{CS}'$

Composition Theorem:

If $\mathcal{CS}$ & $\mathcal{CS}'$ are ε- & δ-binding, then $\mathcal{CS} * \mathcal{CS}'$ is $(\varepsilon+\delta)$-binding.

Proof Sketch



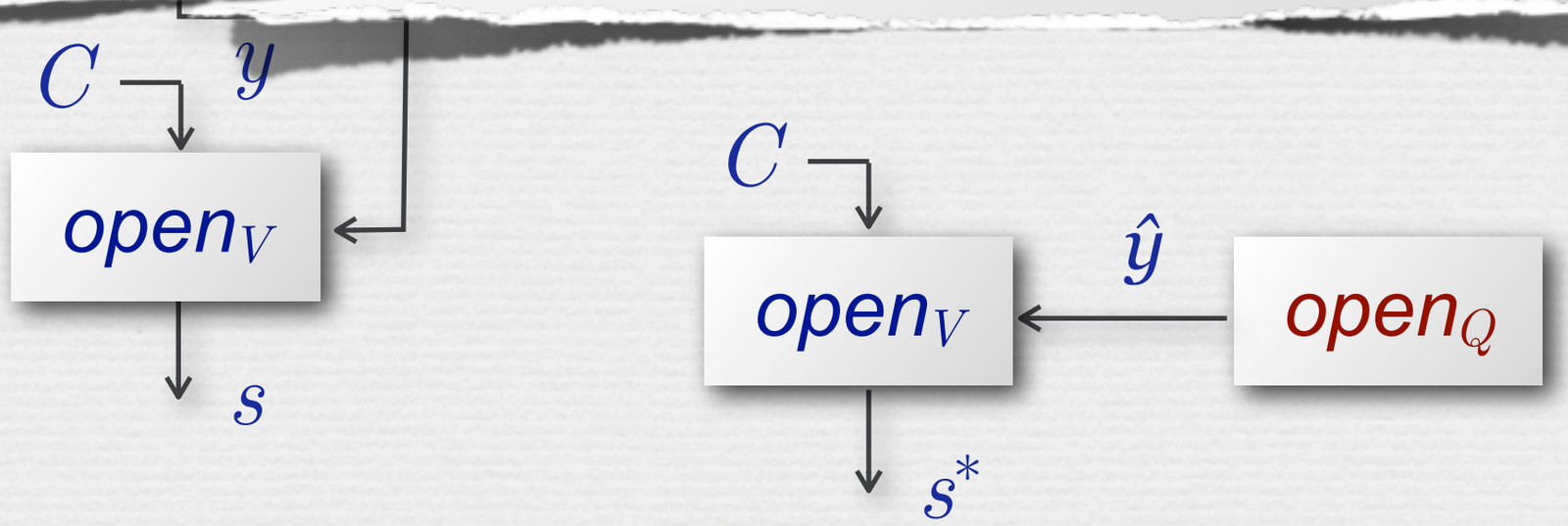
To show:

$$\exists \hat{s} : s \approx_{\varepsilon+\delta} \hat{s} \text{ (or } s = \perp)$$

Proof:

$$\Rightarrow s = open_V(C, y) \approx_\delta open_V(C, \hat{y}) = s^* \approx_\varepsilon \hat{s}$$

Formal proof is more subtle: requires to "glue together" certain distributions, and the argument fails in the quantum setting



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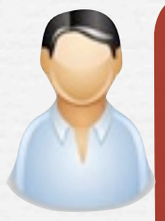
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(Almost) Done?

commitment to b



$a_0 \in \mathbb{F}_2^n$

$x_0 := y_0 \oplus a_0 b$

syntactically: commitment to y_1

$\mathbb{F}_2^n$

$x_1 := y_1 \oplus a_1 y_0$

syntactically: commitment to y_0

$a_2 \in \mathbb{F}_2^n$

x_2

However: it is **not** binding!!!



Only thing left to be shown:

Crépeau et al. **string** commitment scheme is binding

How to Deal with This

- Introduce **relaxed** binding property: **fairly binding** ( ): allow $\hat{s} \neq s \neq \perp$, but then provers have **no control** over s
- Show **Crépeau et al. scheme** $2^{-n/2}$-fairly-binding ( )
- Extend **composition theorem** to this relaxed version ( )
- Conclude **security of Lunghi et al.** m-round scheme as a **fairly binding** string commitment scheme ( )
- Show: **fairly binding** $\Rightarrow$ **binding** when $s \in \{0,1\}$ ( )

Corollary. Lunghi et al. m-round scheme is ε-binding with

$$\varepsilon \lesssim m \cdot 2^{-\frac{n}{2}}$$

Summary

- **Improved analysis** of Lunghi et al. scheme:
  - **linear** instead of **doubly exponential** blow-up
  - 10^{50} years instead of 2 ms
- On the way:
  - **better understanding** of binding property
  - general **composition theorem(s)**
- Open problems:
  - composition for a **larger class of schemes**
  - security for a **stronger notion** of binding
  - security against **quantum attacks**





THANK YOU